

List of Publications (As of September 2026)

2027

- **A. Sudhakaran**, C. Singh, N.K. Chaudhari, A.H. Jadhav, Topical advancement in electrochemical conversion of carbon dioxide to methane: recent advances, challenges, and prospects for carbon neutrality, *Fuel* 430 (2027) 141466. [DOI: 10.1016/J.FUEL.2026.141466]

2026

- **A. Sudhakaran**, D. Prasad, M. Saxena, A.K. Samal, G. Gadikota, A.H. Jadhav, Advances in electrochemical CO₂ reduction to CO for enabling A sustainable energy economy, *Prog. Mater. Sci.* 159 (2026) 101655. [DOI: 10.1016/J.PMATSCI.2026.101655]

2025

- S.C. Alla, **A. Sudhakaran**, O.B. Bembalge, A.K. Samal, R.E. Torrejos, M.N.F. Norrrahim, A.H. Jadhav, Deep Eutectic Solvent Engineered Dendritic Fibrous Nano-Silica Catalyst for Sustainable Fixation of CO₂ Into Value-Added Product at Atmospheric Condition, *Appl. Organomet. Chem.* 39 (2025) e7910. [DOI: 10.1002/aoc.7910]
- H. Manjunath, C. Singh, **A. Sudhakaran**, R.E. Torrejos, M.N.F. Norrrahim, A.K. Samal, A.H. Jadhav, Unraveling the in-situ engineered mixed metal oxides anchored on activated carbon cloth as a flexible electrode material for electrochemical reduction of CO₂ into formic acid, *Sustainable Materials and Technologies* 43 (2025) e01260. [DOI: 10.1016/J.SUSMAT.2025.E01260]
- D.B. Bawiskar, C.B. Sherin Mol, **A. Sudhakaran**, V. V. Narayan, N.K. Chaudhari, B.L. Nikam, A.H. Jadhav, An Efficient and Selective Synthesis of Chalcones via Claisen-Schmidt Condensation Reaction Catalyzed by CuO@rGO Nano Catalyst, *Catalysis Letters* 155 (2025) 327–. [DOI: 10.1007/S10562-025-05158-7]
- S.M. Sherin, **A. Sudhakaran**, H. Manjunath, A. Pandey, A.K. Samal, M.N. Mary, A.H. Jadhav, Unlocking the potential of in-situ thermally reduced graphene oxide anchored with ternary metal oxide-spinel catalyst for sustainable high-performance electrochemical valorisation of CO₂, *Fuel* 387 (2025) 134458. [DOI: 10.1016/J.FUEL.2025.134458]

- **A. Sudhakaran**, C. Singh, H.M. Aaradhya, A. Biradar, A.K. Samal, N.K. Chaudhari, A.H. Jadhav, Advancements and Perspective of Environmentally Sustainable Technologies for Electrochemical Selective Conversion of CO₂ to Methanol, *Catalysis Reviews* 67 (2025) 845–957. [DOI: 10.1080/01614940.2024.2340582]
- N.A. Raju, S.C. Alla, **A. Sudhakaran**, R.E. Torrejos, M.N.F. Norrrahim, A.K. Samal, A.H. Jadhav, In-Situ MgO engineered dendritic fibrous nano silica as an efficient catalyst for fixation of CO₂ at atmospheric pressure, *Chem. Eng. Sci.* 317 (2025) 122068. [DOI: 10.1016/J.CES.2025.122068]

2024

- **A. Sudhakaran**, A. Sudhakaran, E. Sivasenthil, Structure-property connections in dual-phase multifunctional barium titanate based nanocomposites: Insights from charge density distributions, *Mater. Chem. Phys.* 318 (2024) 129290. [DOI: 10.1016/J.MATCHEMPHYS.2024.129290]
- **D. Rajanna**, **P.M. Srinivasappa**, **A. Sudhakaran**, A. Biradar, A.K. Samal, O.B. Bembalge, A.H. Jadhav, Recent Advances in the Fixation of CO₂ into Quinazoline and Benzimidazole, *Energy & Fuels* 38 (2024) 8481–8515. [DOI: 10.1021/ACS.ENERGYFUELS.4C00277]
- C. Singh, H. Manjunath, **A. Sudhakaran**, O.B. Bembalge, R. Eliseo Torrejos, M. Nor Faiz Norrrahim, A.H. Jadhav, Unleashing the potential of N-doped metal-free bi-functional hollow carbon spheres: Catalyst for environmental conservation and advancing sustainable energy solutions, *Fuel* 374 (2024) 132457. [DOI: 10.1016/J.FUEL.2024.132457]
- R.S. Mohammed, **A. Sudhakaran**, M.Y.A. Mostafa, G. Abbady, Synthesize of ZnO and CuO nanoparticles with plasma jet at different treatment times and testing its optical parameters with UV-Vis-NIR, *Appl. Phys. A Mater. Sci. Process.* 130 (2024) 533–. [DOI: 10.1007/s00339-024-07651-z]
- **A. Sudhakaran**, A. Sudhakaran, E. Sivasenthil, Comprehensive Analysis of Optical, Dielectric, and Electrical Properties in Al- and Cu-Doped Barium Hexaferrite/Cobalt–Zinc Ferrite Hybrid Nanocomposites, *Journal of Electronic Materials* 53 (2024) 4622–4644. [DOI: 10.1007/S11664-024-11197-4]
- W.R. Talib, **A. Sudhakaran**, A. Sudhakaran, R.S. Mohammed, Potential biological and optoelectronic applications of AgO:ZnO nanocomposite synthesized by green

approach, *Eur. Phys. J. Plus* 139 (2024) 1118. [DOI: 10.1140/epjp/s13360-024-05920-7]

- X. Li, J. Zhao, Y. Zhu, R.S. Ahmed, R.S. Mohammed, A. Mustafa, A. Majeed, **A. Sudhakaran**, Biological activity of MgO nanoparticle synthesis by plasma-assisted reduction method, *Phys. Scr.* 99 (2024) 115901. [DOI: 10.1088/1402-4896/ad7dbc]
- **A. Sudhakaran**, A. Sudhakaran, A Topological Control Algorithm for Decision-Making in Complex Material Systems (2024). www.tnsroindia.org.in.

2023

- A. Sudhakaran, **A. Sudhakaran**, E. Sivasenthil, Electron density effect of hybrid ferrites on structural, spectroscopic, optical, and magnetic properties in (BaTiO₃) (1-x) + (ZnFe₂O₄) x nanoceramics, *Journal of Materials Science: Materials in Electronics* 34 (2023) 104–. [DOI: 10.1007/S10854-022-09563-1]
- **A. Sudhakaran**, A. Sudhakaran, E. Sivasenthil, Investigating structure, elasticity, morphology, composition, and optical behavior of Al-doped BaFe₁₂O₁₉/CoZnFe₂O₄ hybrid composite, *Journal of Materials Research* 38 (2023) 1239–1253. [DOI: 10.1557/S43578-022-00879-6]
- **A. Sudhakaran**, A. Sudhakaran, S. E, Multifunctionality of AlBaFe₁₂O₁₉/CoZnFe₂O₄ hybrid nanocomposite: Promising structural, elastic, morphological, compositional, optical, and magnetic properties, *Journal of Physics and Chemistry of Solids* 174 (2023) 111134. [DOI: 10.1016/J.JPCS.2022.111134]
- **A. Sudhakaran**, E. Sivasenthil, Ferrites: A study on structure, properties, synthesis, characterization, and Applications, *Journal of Xidian University* 17(7) (2023). [DOI: 10.37896/jxu17.7/122]

2022

- **A. Sudhakaran**, A. Sudhakaran, E. Siva Senthil, Optical Band gap characterization of Novel Nanoscale CBM-CZF composite, *Env. Pharmacol. Life Sci.* (2022) 643–650. [DOI: 10.5281/zenodo.7103715]
- A. Sudhakaran, **A. Sudhakaran**, E. Sivasenthil, Novel Heterostructure Ba₂TiO₃-Zn_xFe₂O₄ Nanocomposite: A Quantitative Study, *Rasayan J. Chem* 15 (2022) 2197–2202. [DOI: 10.31788/RJC.2022.1546964]

- A. Sudhakaran, **A. Sudhakaran**, E. Siva Senthil, Qualitative Analysis of Magnetic Moment in $\text{Ba}_x\text{TiO}_3 - \text{Ni}_x\text{Fe}_{3-x}\text{O}_4$ Nanocomposite Synthesized by Novel low Temperature Technique, *International Journal of Recent Technology and Engineering (IJRTE)* 10 (2022) 17–22 / 2277–3878. [DOI: 10.35940/ijrte.D6628.0110522]
- **A. Sudhakaran**, A. Sudhakaran, E. Sivasenthil, Investigating Optical Properties of Novel Heterostructure Aluminium Doped Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) Nanocomposite, *Indian Journal of Natural Sciences* 13 (2022). [DOI: 10.5281/zenodo.6945268]
- **A. Sudhakaran**, A. Sudhakaran, E. Sivasenthil, Deliberating Structural, Optical and Spectroscopic Properties of Novel Barium Titanate Zinc Ferrite Nanocomposite Synthesized Using Low Temperature Techniques, *Stochastic Modeling & Applications* 26(3) (June 2022).
- **A. Sudhakaran**, A. Sudhakaran, E. Sivasenthil, Studying Structural and Optical Properties of Novel Aluminium Barium Hexaferrite (ALBAM) Cobalt Zinc Ferrite (CZFO) nanocomposite, *Stochastic Modeling & Applications* 26(3) (June 2022).

2021

- E.S. Senthil, R. Yogeshwaran, B.S. Yeshwanth, S.S. Banu, **A. Sudhakaran**, A. Sudhakaran, Solar Energy Based LED Powered Light and Inverter Circuit with Different Types of Sensors, *IOP Conf. Ser. Mater. Sci. Eng.* 1126 (2021) 012029. [DOI: 10.1088/1757-899X/1126/1/012029]
- **A. Sudhakaran**, A. Sudhakaran, E. Siva Senthil, Study of Bandgap Energy of Novel $(\text{Cu}_{0.5}\text{Ba}_{0.5}\text{Fe}_{12}\text{O}_{19})_{1-x} / (\text{Co}_{0.6}\text{Zn}_{0.4}\text{Fe}_2\text{O}_4)_x$ Nanocomposite, *International Journal of Recent Technology and Engineering* (2021). [DOI: 10.35940/ijrte.D6607.1110421]